Indian Institute of Information Technology, Design and Manufacturing, Kurnool

Department of Electronics and Communication Engineering

4-Bit Magnitude Comparator

Full-Custom Design in Cadence Virtuoso

Schematic, Simulation, Layout and Physical Verification in 180 nm CMOS (gpdk180)

Project Report

Course: **Digital IC Design Practice**

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Abstract

This report presents the complete full-custom design of a 4-bit magnitude comparator using Cadence Virtuoso and the 180 nm generic process design kit (gpdk180). The comparator accepts two 4-bit unsigned binary numbers $A[3:0]$ and $B[3:0]$ and produces three mutually exclusive outputs indicating whether $A > B$, $A = B$ or $A < B$.

The design is built hierarchically. Basic CMOS cells (inverter, AND2, OR2 and XNOR2) are first designed at transistor level, each with a schematic, a symbol and a hand-drawn layout. These cells are combined into a 1-bit comparator, and four such stages together with combining logic form the 4-bit comparator. All transistors use a width of 2 μm and the minimum channel length of 180 nm and operate from a 1.8 V supply. Functional behaviour was examined through transient simulation with the Spectre simulator, and every layout was verified with the Design Rule Check (DRC), which reported no errors for all cells and for the 1-bit and 4-bit comparators.

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1 4-Bit Magnitude Comparator

1.1 Background

Comparing two numbers is one of the most basic operations in digital systems. Arithmetic logic units use it for conditional branching, memories and caches use it for address and tag matching, sorting hardware uses it to order data, and control systems use it to compare a measured value against a threshold. A *magnitude comparator* is the combinational circuit that performs this operation in hardware.

In modern integrated circuits, most digital blocks are produced by automated synthesis and place-and-route tools. A *full-custom* approach, in contrast, gives the designer direct control over every transistor and every polygon of the layout. This leads to compact and well-tuned circuits and, equally important, gives a clear understanding of how a logic function is turned into silicon.

1.2 Objectives

The objectives of this project are:

1. to design a 4-bit magnitude comparator at transistor level using static CMOS logic;
2. to build the design hierarchically from reusable cells (INV, AND2, OR2, XNOR2);
3. to verify the functionality by transient simulation in Cadence Virtuoso (ADE and Spectre);
4. to draw the full-custom layout of every cell and of the complete comparator in the gpdk180 technology; and
5. to verify each layout with the Design Rule Check (DRC).

1.3 Tools and Technology

Table 1. Tools and technology used

Item	Details
Design environment	Cadence Virtuoso 6.1.8 (Schematic Editor, Layout Suite XL)
Simulator	Cadence Spectre (ADE), Visualization & Analysis XL
Verification	Design Rule Check (DRC)
Technology	gpdk180 (180 nm generic CMOS PDK)
Supply voltage	$V_{DD} = 1.8\,\mathrm{V}$
Transistor sizes	$W = 2\,\mu\mathrm{m}$, $L = 180\,\mathrm{nm}$ (NMOS and PMOS)

1.4 Circuit Diagram

The overall circuit diagram of the 4-bit magnitude comparator is shown in Fig. 1. It shows how the bit-wise comparison stages and the combining logic are connected to produce the outputs $A > B$, $A = B$ and $A < B$.

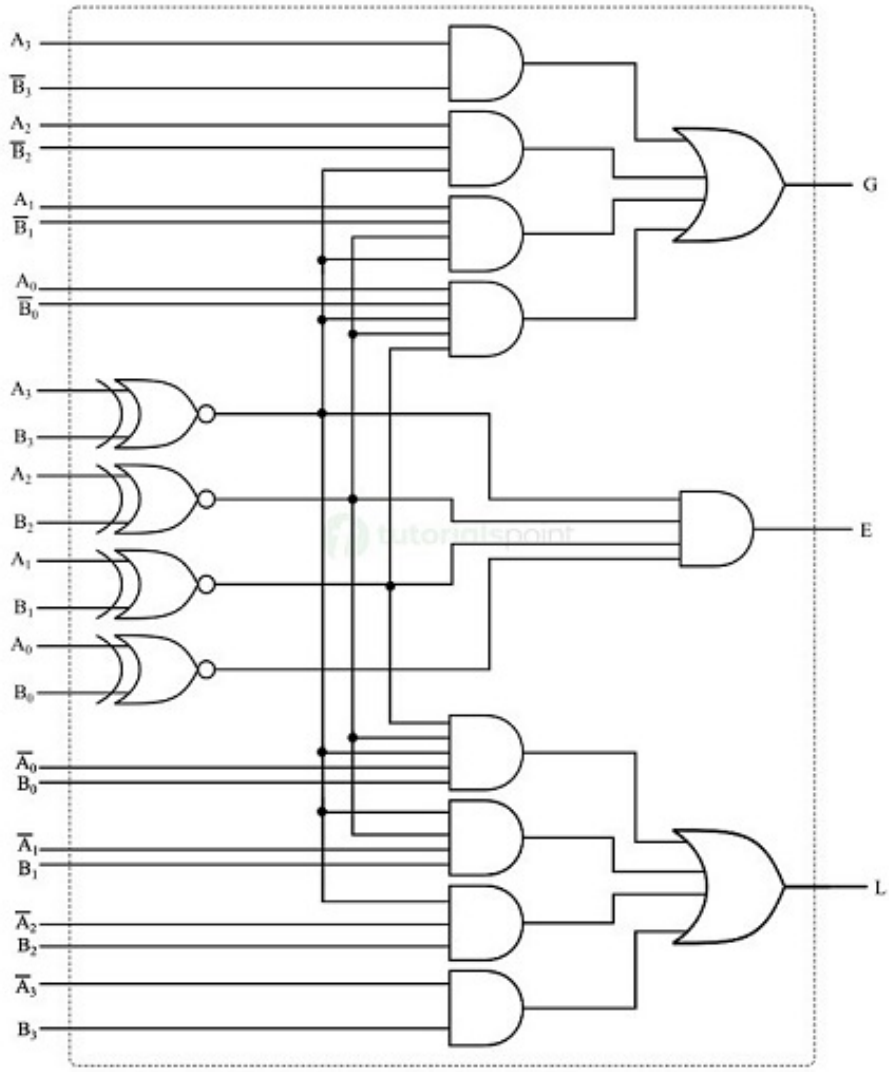


Figure 1. Circuit diagram of the 4-bit magnitude comparator

1.5 Organisation of the Report

Section 2 presents the theory of magnitude comparison and of static CMOS logic. Section 3 describes the design methodology and the cell hierarchy. Section 4 presents the transistor-level schematics, Section 5 the simulation results, Section 6 the layouts and Section 7 the physical verification. Section 8 summarises the results, and Section 9 concludes the report and lists possible future work.

2 Theory and Background

2.1 Magnitude Comparator

A magnitude comparator is a combinational circuit that compares two n-bit unsigned binary numbers A and B and produces three outputs: $A > B$, $A = B$ and $A < B$. For every input combination exactly one output is logic 1, that is,

$$(A > B) + (A = B) + (A < B) = 1. \tag{1}$$

Since the circuit is combinational, the outputs depend only on the present inputs and no clock or memory element is required.

2.2 Bit-by-Bit Comparison

Two binary numbers are compared starting at the most significant bit (MSB), exactly as decimal numbers are compared digit by digit:

1. If $A_3 \neq B_3$, the number having the 1 in this position is larger and the comparison is finished.
2. If $A_3 = B_3$, the next lower bit pair (A_2, B_2) is examined, and so on.
3. If all four bit pairs are equal, then $A = B$.

For example, with $A = 1010_2$ (ten) and $B = 1001_2$ (nine), the two upper bits are equal, while $A_1 = 1$ and $B_1 = 0$; therefore $A > B$.

2.3 One-Bit Comparator

For a single bit pair the truth table is given in Table 2.

Table 2. Truth table of the 1-bit comparator

A	B	$A > B$ (GT)	$A = B$ (EQ)	$A < B$ (LT)
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

From the truth table,

$$G = A \cdot \overline{B}, \tag{2}$$

$$L = \overline{A} \cdot B, \tag{3}$$

$$E = A \odot B = A\,B + \overline{A}\,\overline{B}. \tag{4}$$

A 1-bit comparator therefore needs one XNOR gate, two inverters and two AND gates, which is exactly the structure used in this project (Section 4.5).

2.4 Four-Bit Comparator

For each bit position i the three signals $G_i = A_i\overline{B_i}$, $L_i = \overline{A_i}B_i$ and $E_i = A_i \odot B_i$ are generated. The 4-bit outputs are then

$$(A = B) = E_3\,E_2\,E_1\,E_0, \tag{5}$$

$$(A > B) = G_3 + E_3G_2 + E_3E_2G_1 + E_3E_2E_1G_0, \tag{6}$$

$$(A < B) = L_3 + E_3L_2 + E_3E_2L_1 + E_3E_2E_1L_0. \tag{7}$$

The term E_3G_2, for instance, means that the MSBs are equal *and* bit 2 of A is greater than bit 2 of B. A lower bit pair matters only if all higher bit pairs are equal, so the equality terms are chained in order of significance. Table 3 lists a few representative cases.

Table 3. Representative 4-bit comparison cases

A	B	Decided at	$A > B$	$A = B$	$A < B$
0000	0000	all bits equal	0	1	0
1000	0111	bit 3 ($G_3 = 1$)	1	0	0
0111	1000	bit 3 ($L_3 = 1$)	0	0	1
1100	1010	bit 2 ($E_3 = 1,\ G_2 = 1$)	1	0	0
0101	0110	bit 1 ($E_3E_2 = 1,\ L_1 = 1$)	0	0	1
1011	1010	bit 0 ($E_3E_2E_1 = 1,\ G_0 = 1$)	1	0	0
1111	1111	all bits equal	0	1	0

The worst-case delay occurs when the decision depends on the least significant bit, because the signal then has to travel through the XNOR gate, the chain of AND gates and the final OR gate.

2.5 Static CMOS Logic

All cells in this design are static CMOS gates. Each gate consists of a pull-up network (PUN) of PMOS transistors connected to V_{DD} and a complementary pull-down network (PDN) of NMOS transistors connected to ground. A PMOS transistor conducts when its gate is low, an NMOS transistor conducts when its gate is high, and for every input combination exactly one of the two networks conducts. This gives full-swing outputs, very low static power and high noise margins.

- **Inverter:** one PMOS and one NMOS, $Y = \overline{A}$.
- **NAND2:** NMOS transistors in series, PMOS transistors in parallel, $Y = \overline{A \cdot B}$.
- **NOR2:** NMOS transistors in parallel, PMOS transistors in series, $Y = \overline{A + B}$.
- **AND2 and OR2:** a NAND2 or NOR2 followed by an inverter, which restores the polarity and gives a buffered output.
- **XNOR2:** $Y = A\,B + \overline{A}\,\overline{B}$, used here as the bit-equality detector.

2.6 Performance Metrics

- **Propagation delay** t_{pd}: the time from the 50 % point of an input transition to the 50 % point of the resulting output transition.
- **Power**: the dynamic power $P_{dyn} = \alpha C V_{DD}^2 f$ plus the (small) leakage power.
- **Noise margins**: $NM_H = V_{OH} - V_{IH}$ and $NM_L = V_{IL} - V_{OL}$.
- **Power-delay product**: $P \times t_{pd}$, a figure of merit for the energy per operation.

2.7 Full-Custom Design Flow

In a full-custom flow every transistor is sized and every layout polygon is drawn manually, as opposed to a semi-custom (standard-cell) flow in which synthesis and automatic place-and-route tools use a ready-made cell library. The full-custom approach takes more effort but gives the best control over area, speed and power, and is typically used for analog blocks, memories and speed-critical datapaths. The flow followed here is shown in Fig. 2.

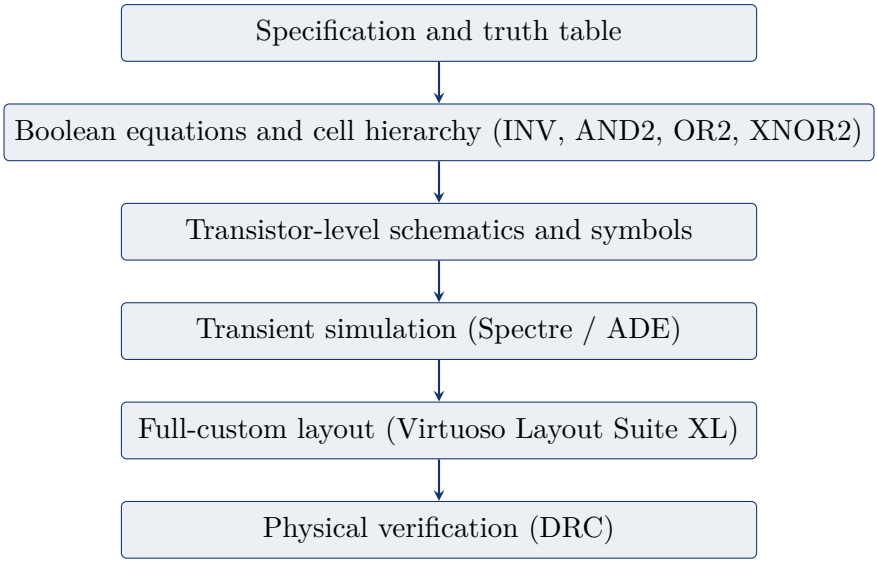


Figure 2. Full-custom design flow followed in this project

2.8 Layout and Verification Concepts

- **Layers (gpdk180):** N-well, active area (oxide), polysilicon (gates), n+ and p+ implants, contacts, Metal1 to Metal6 and the vias between them. The drawn gate length of this process is 180 nm.
- **Design rules:** minimum width, spacing, enclosure and extension rules specified by the foundry for every layer.
- **Well and substrate contacts:** the N-well is tied to V_{DD} and the p-substrate to ground to bias the transistors correctly and to avoid latch-up.
- **Routing:** Metal1 is used for local connections inside a cell, higher metals for inter-cell routing, and polysilicon for gates and short local connections.
- **DRC** checks that the drawn layout obeys all geometric design rules.

3 Design Methodology

3.1 Cell Hierarchy

The comparator is built bottom-up in four levels (Table 4). Each cell has a schematic, a symbol and a layout view in its own library, so that it can be instantiated in the next level up.

Table 4. Hierarchy of the design

Level	Cell	Description
1	Inverter (INV)	One PMOS and one NMOS transistor
2	AND2	NAND2 (4 transistors) followed by an inverter
2	OR2	NOR2 (4 transistors) followed by an inverter
2	XNOR2	Eight transistors with two input inverters (bit-equality)
3	1-bit comparator	One XNOR2, two inverters and two AND2 gates
4	4-bit comparator	Four 1-bit comparator stages with combining AND2/OR2 logic

3.2 Transistor Sizing

All transistors in the design use $W = 2\,\mu$m and $L = 180\,$nm with multiplier $m = 1$, taken from the `nmos1` and `pmos1` devices of the gpdk180 library. Using a common size for all devices keeps the cells uniform and simplifies the layout. The sizing is therefore not individually optimised for delay; optimising it (for example $W_p \approx 2W_n$ for balanced rise and fall times) is mentioned under future work.

Table 5. Device parameters used in all cells

Device	Library cell	W	L
NMOS	`nmos1`	$2\,\mu$m	180 nm
PMOS	`pmos1`	$2\,\mu$m	180 nm

4 Schematic Design

4.1 Inverter

The inverter (Fig. 3) consists of one PMOS transistor connected between V_{DD} and the output and one NMOS transistor connected between the output and ground, with both gates joined to the input. It has 2 instances and implements $Y = \overline{A}$.

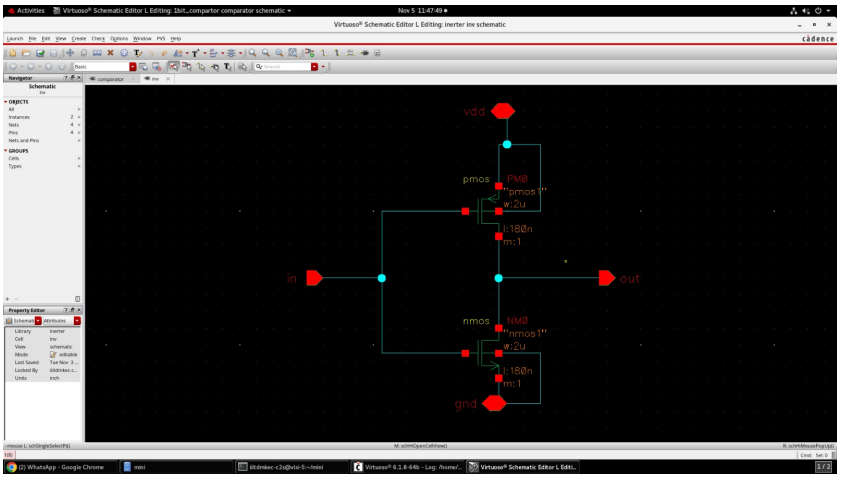


Figure 3. Transistor-level schematic of the inverter

4.2 AND2 Gate

The AND2 gate (Fig. 4) is a CMOS NAND2 stage made of two parallel PMOS and two series NMOS transistors, followed by the inverter symbol of the previous section. The inverter restores the polarity and drives the output.

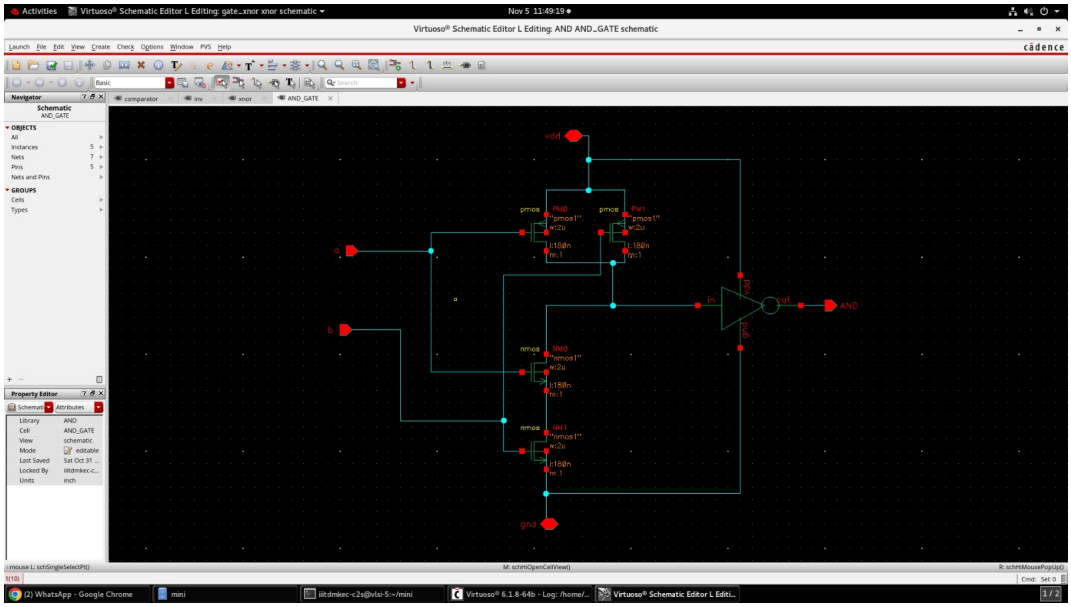


Figure 4. Transistor-level schematic of the AND2 gate

4.3 OR2 Gate

The OR2 gate (Fig. 5) is a CMOS NOR2 stage with two series PMOS and two parallel NMOS transistors, followed by an inverter.

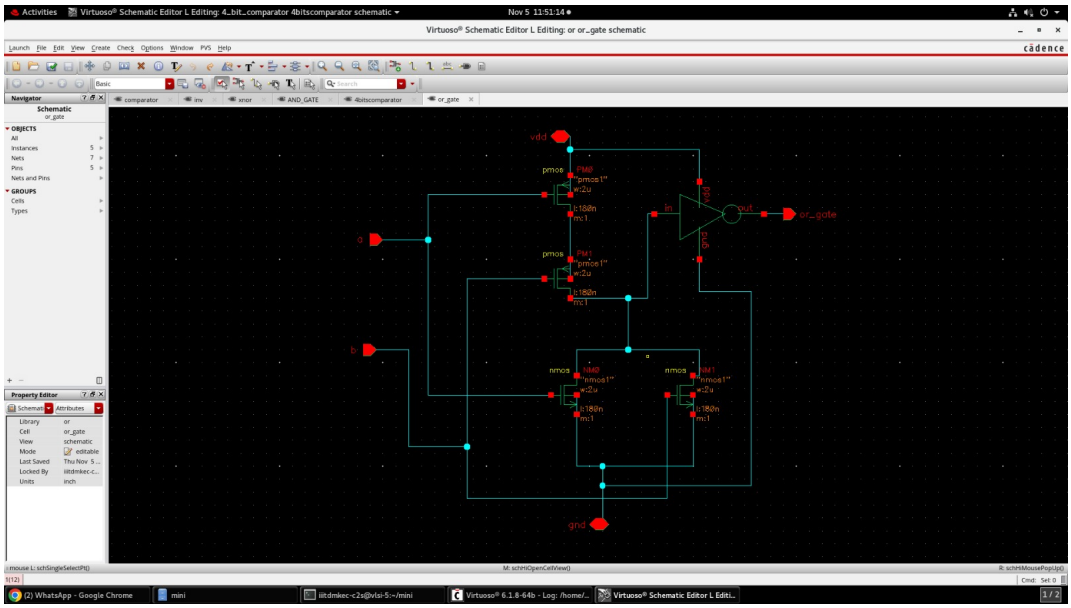


Figure 5. Transistor-level schematic of the OR2 gate

4.4 XNOR2 Gate

The XNOR2 gate (Fig. 6) detects whether its two inputs are equal. It uses two input inverters to generate $\overline{A}$ and $\overline{B}$ and eight transistors (four PMOS and four NMOS), 10

instances in total, and gives the output $Z = A \odot B$.

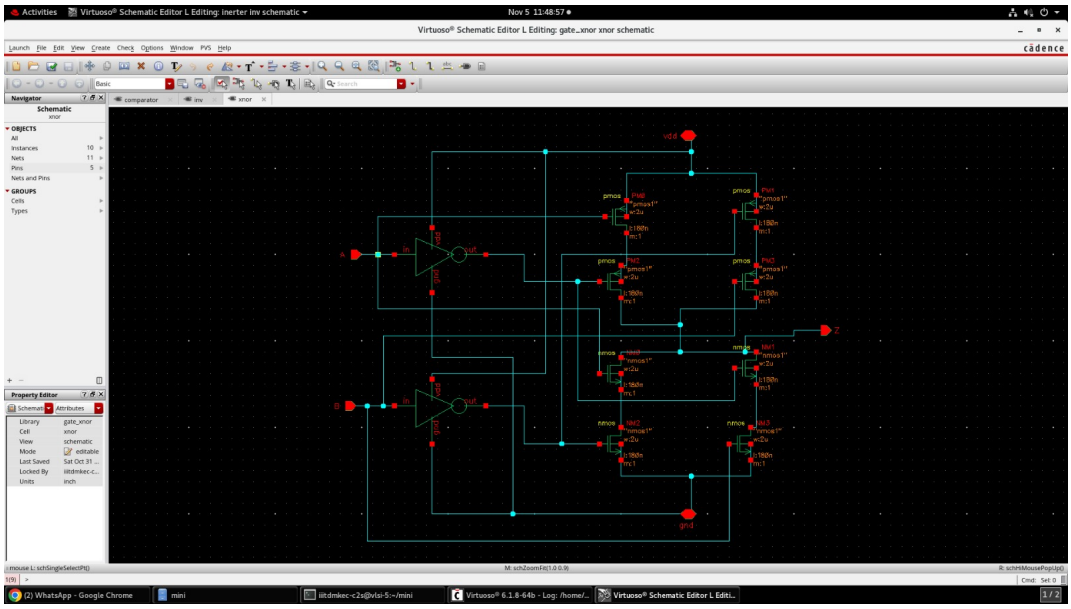


Figure 6. Transistor-level schematic of the XNOR2 gate

4.5 One-Bit Comparator

The 1-bit comparator (Fig. 7) contains five instances: one XNOR2 gate that produces the equality output, two inverters that produce $\overline{A}$ and $\overline{B}$, and two AND2 gates that produce $G = A\overline{B}$ (GT) and $L = \overline{A}B$ (LT). The cell has the inputs A and B and the outputs GT, EQ and LT, together with the supply pins.

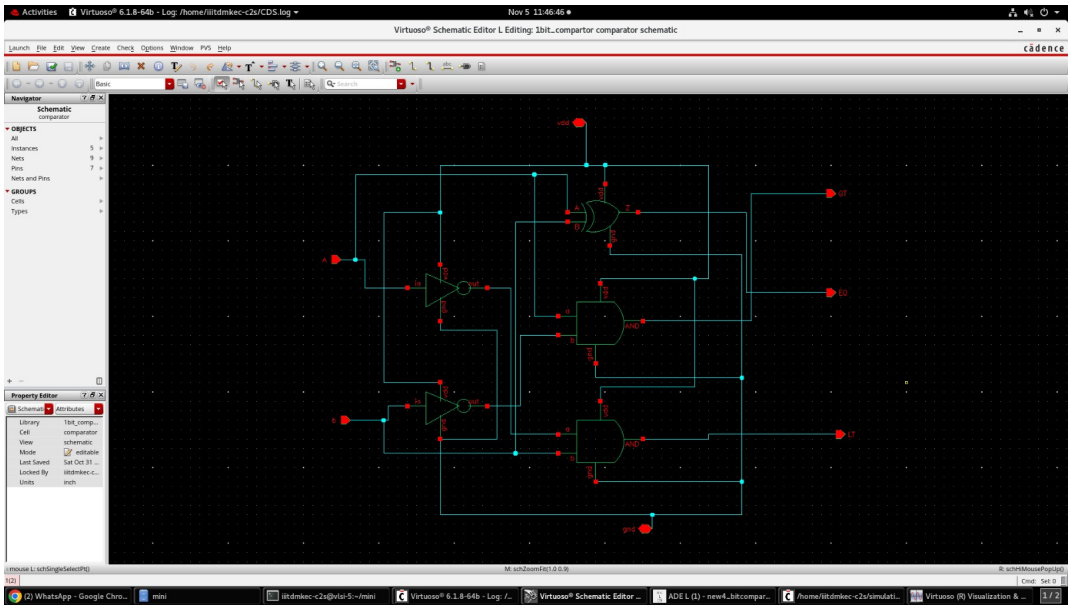


Figure 7. Schematic of the 1-bit comparator

4.6 Four-Bit Comparator

The 4-bit comparator (Fig. 8) is built from four instances of the 1-bit comparator, one for each bit pair, together with AND2 and OR2 gates that implement the chained equations of Section 2. The top-level schematic has 39 instances and 13 pins: the inputs $A_3 \ldots A_0$ and $B_3 \ldots B_0$, the outputs GT, EQ and LT, and the supply pins V_{DD} and GND.

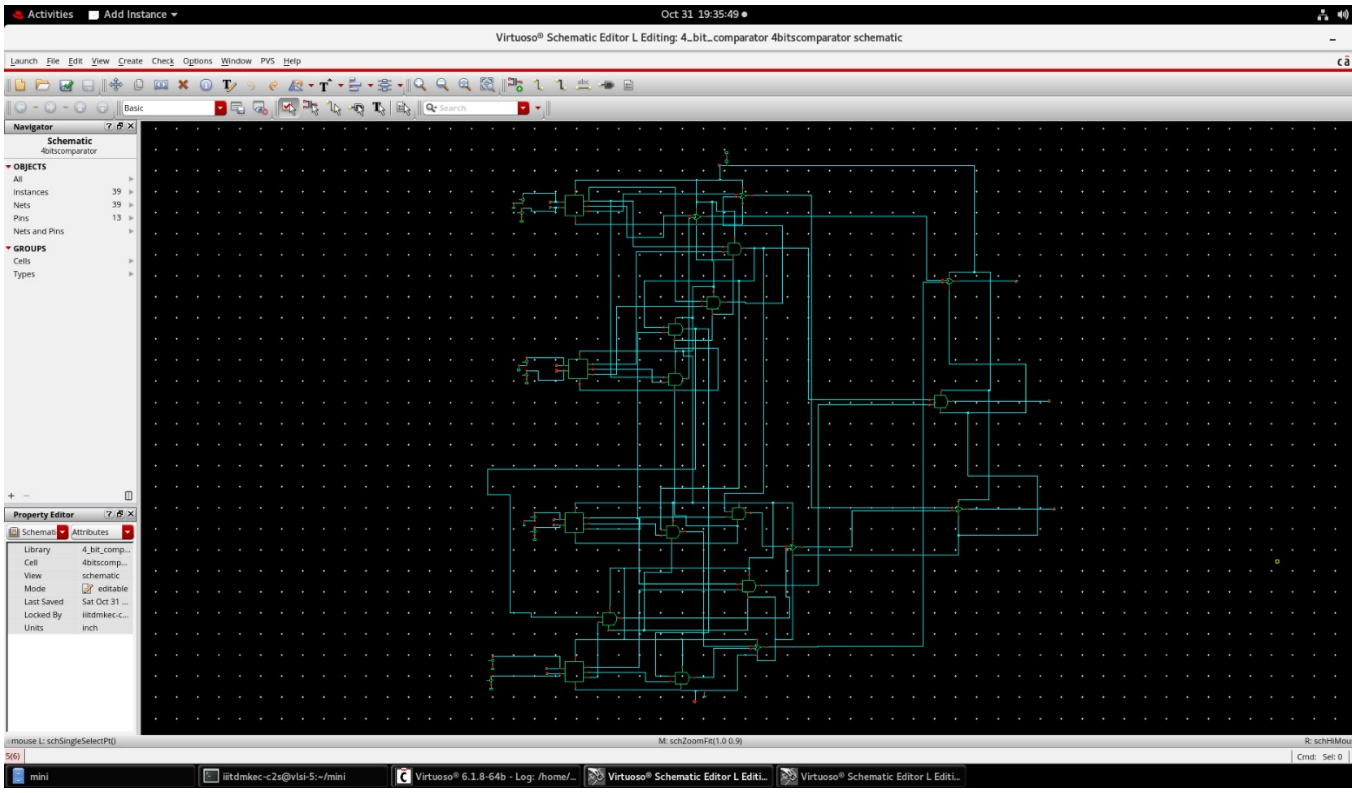


Figure 8. Top-level schematic of the 4-bit magnitude comparator

5 Simulation Results

5.1 Simulation Setup

The circuits were simulated with the Spectre simulator through the Analog Design Environment (ADE) using the gpdk180 device models and a supply of $V_{DD} = 1.8\,\text{V}$. Pulse sources of different periods were applied to the inputs so that every input combination appears, and the transient responses were plotted in the Visualization & Analysis XL tool.

5.2 One-Bit Comparator

The expected behaviour is given by Table 2. The simulated waveforms of the 1-bit comparator over 100 ns are shown in Fig. 9. Input A is high from 0 to 40 ns and again after 80 ns, and input B is high from 0 to 20 ns and after 60 ns. Between 20 and 40 ns, when $A = 1$ and $B = 0$, the GT output goes to the full supply level while the other outputs stay low, as required by the truth table.



Figure 9. Transient simulation of the 1-bit comparator (inputs A, B and outputs GT, EQ, LT)

5.3 Design Summary

Table 6. Summary of the designed cells (from the schematic and layout views)

Cell	Instances	Transistors	Schematic	DRC
Inverter	2	2	Done	Clean
AND2	5	6	Done	Clean
OR2	5	6	Done	Clean
XNOR2	10	12	Done	Clean
1-bit comparator	5	28	Done	Clean
4-bit comparator	39	—	Done	Clean

Transistor counts include the transistors inside the inverter instances of each cell.

6 Layout Design

6.1 Layout Approach

Each cell was laid out manually in Virtuoso Layout Suite XL using the gpdk180 layers. The layouts use Metal1 for local routing inside the cells, higher metal layers and vias for connections between cells, polysilicon for gates, and N-well and substrate contacts connected to V_{DD} and ground rails. Pins were placed for all inputs, outputs and supply connections. The cells were then instantiated in the 1-bit comparator layout, which in turn was instantiated four times in the 4-bit comparator.

6.2 Inverter Layout

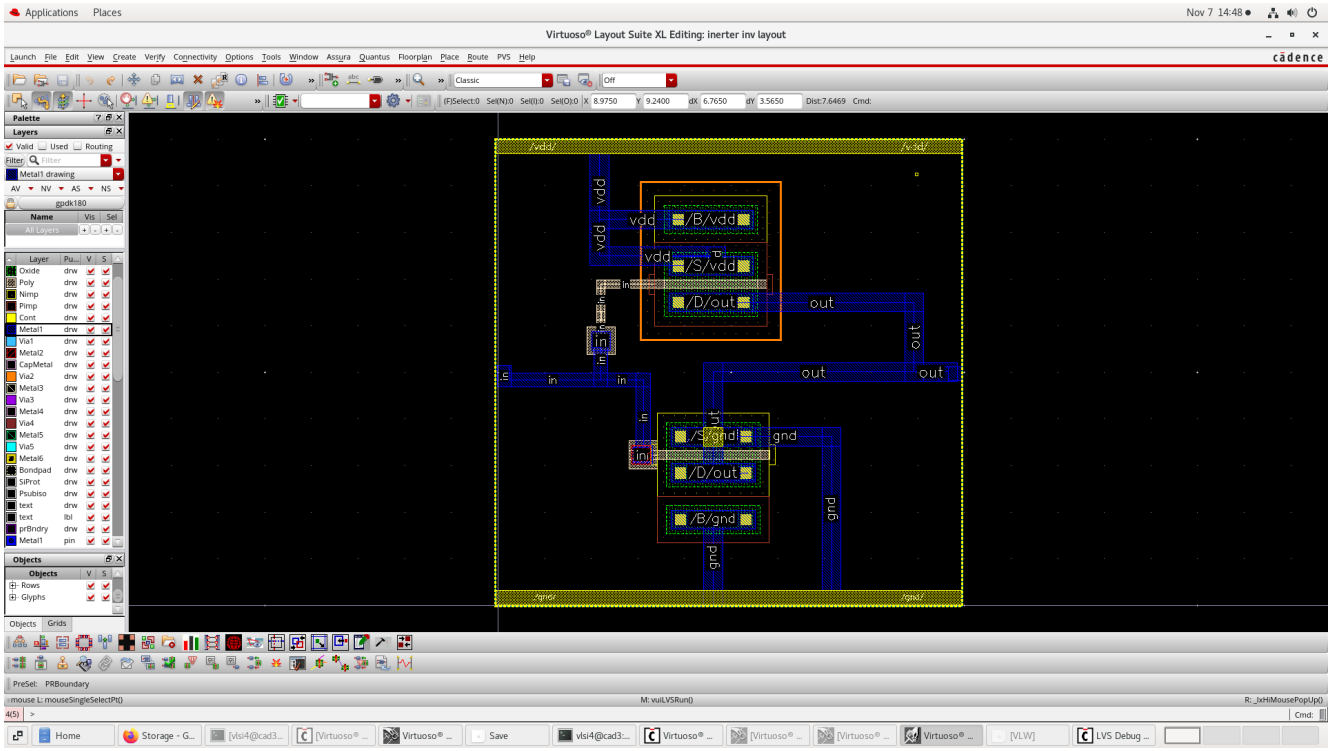


Figure 10. Layout of the inverter

6.3 AND2 Layout

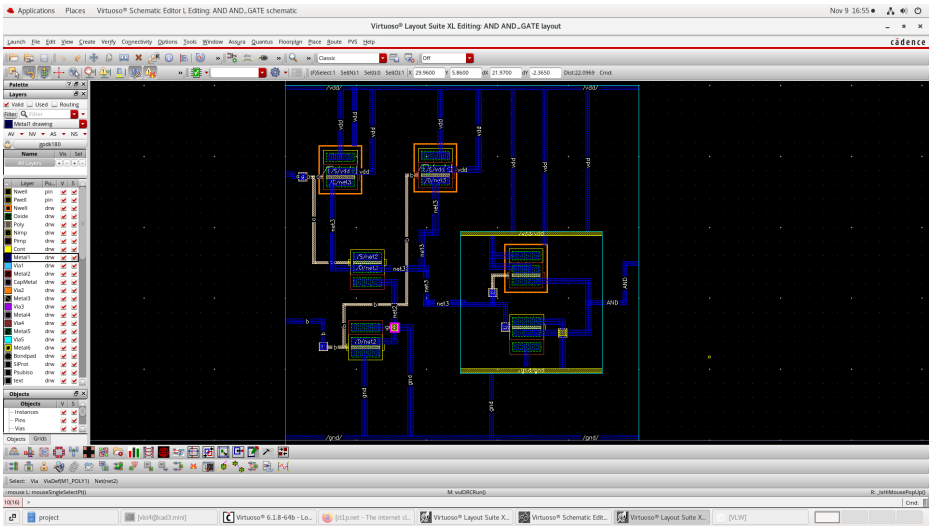


Figure 11. Layout of the AND2 gate

6.4 OR2 Layout

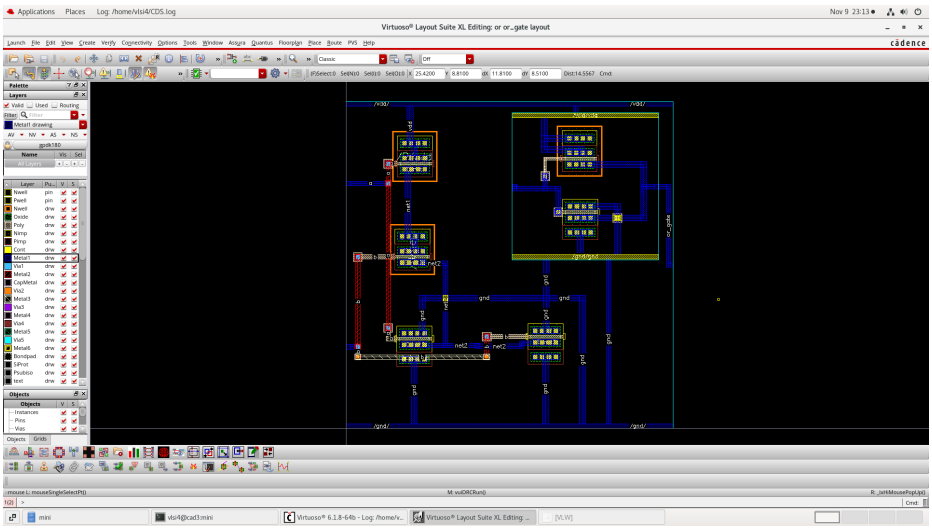


Figure 12. Layout of the OR2 gate

6.5 XNOR2 Layout

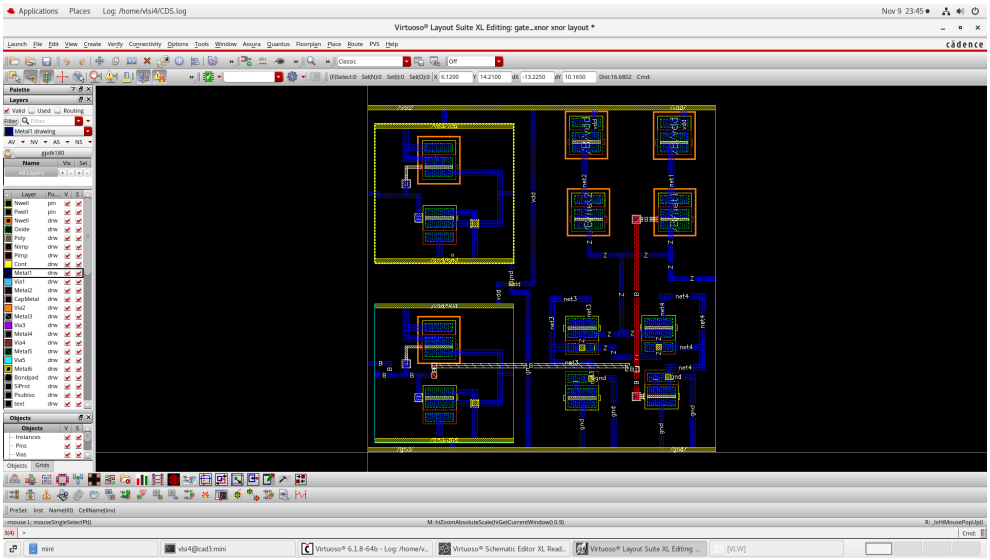


Figure 13. Layout of the XNOR2 gate

6.6 One-Bit Comparator Layout

The 1-bit comparator layout (Fig. 14) places the XNOR2, AND2 and inverter cells inside a common boundary and connects them with Metal1 and Metal2 routing.

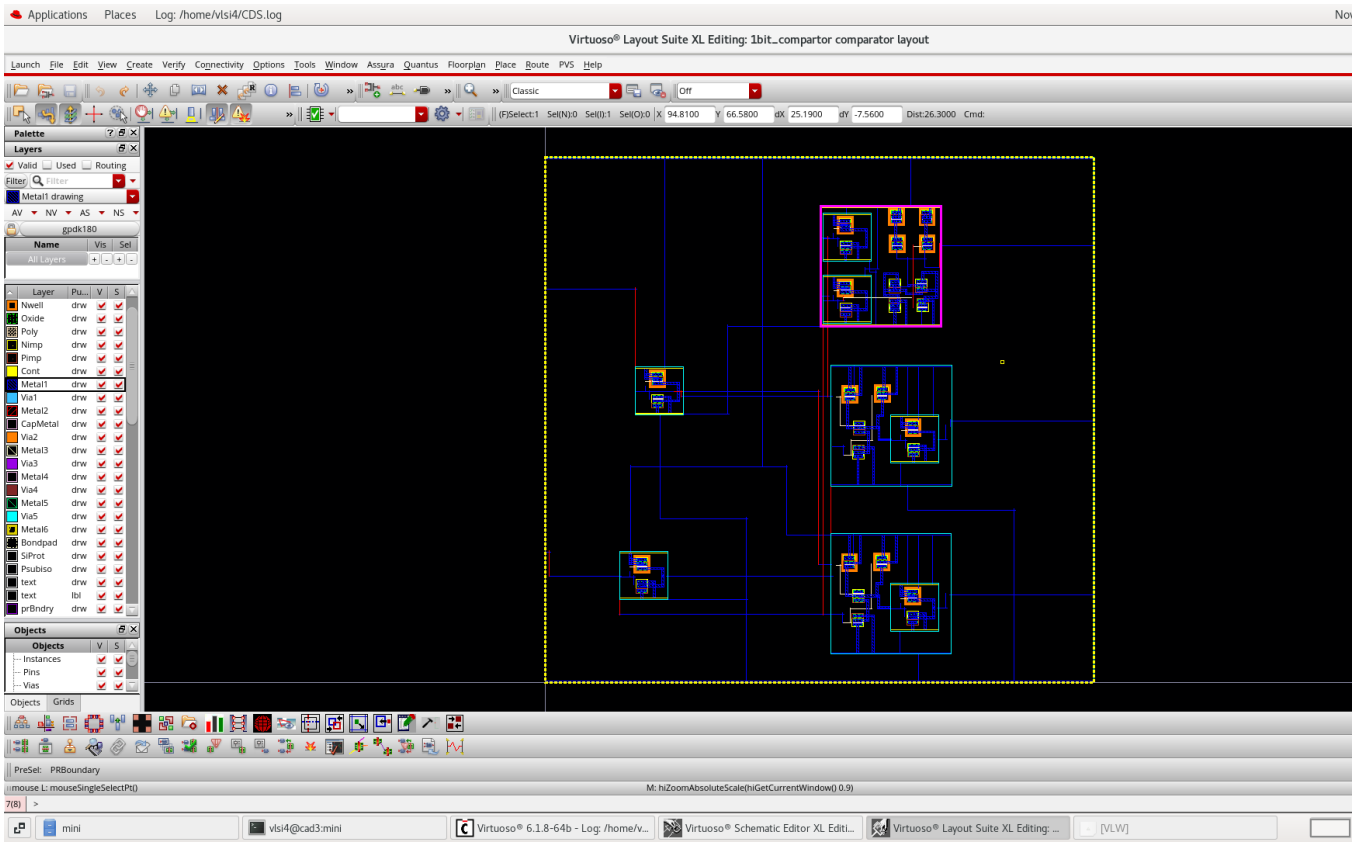


Figure 14. Layout of the 1-bit comparator

6.7 Four-Bit Comparator Layout

The complete 4-bit comparator layout (Fig. 15) contains the four 1-bit comparator blocks and the combining gates, routed on several metal layers.

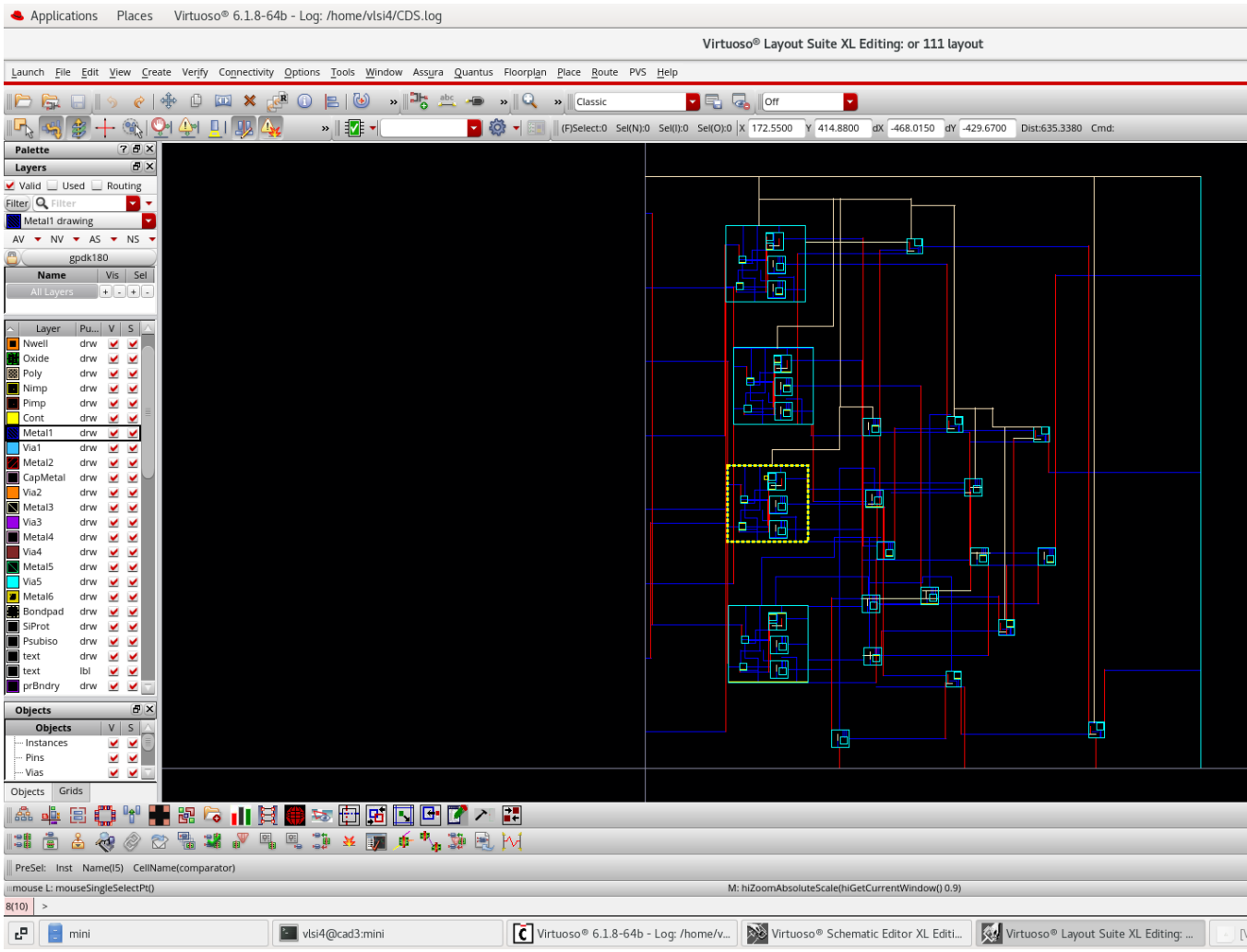
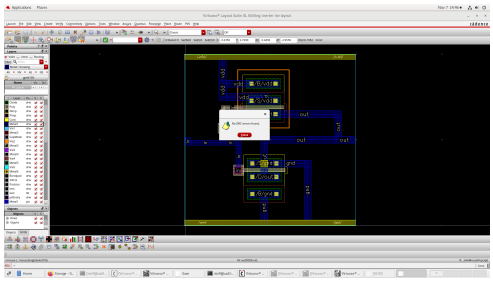


Figure 15. Layout of the 4-bit magnitude comparator

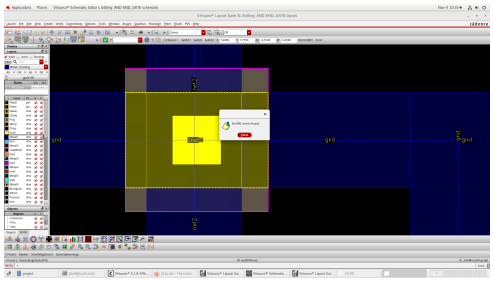
7 Physical Verification

7.1 Design Rule Check

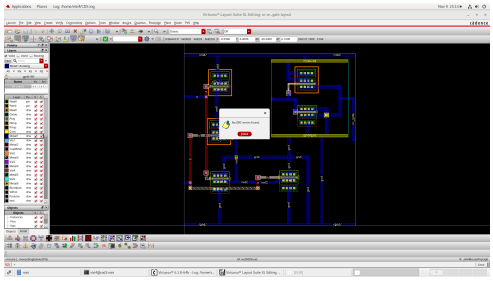
The Design Rule Check verifies that the drawn layout obeys all geometric rules of the process (minimum width, spacing, enclosure and extension). The DRC was run on every layout, and for each of them the tool reported *"No DRC errors found"* (Fig. 16 and Table 7).



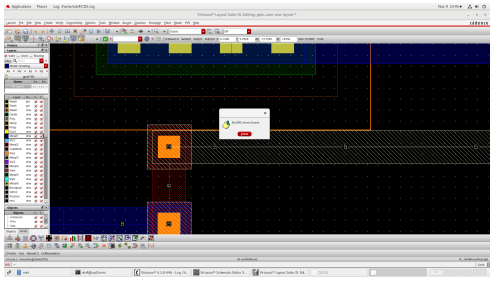
(a) Inverter



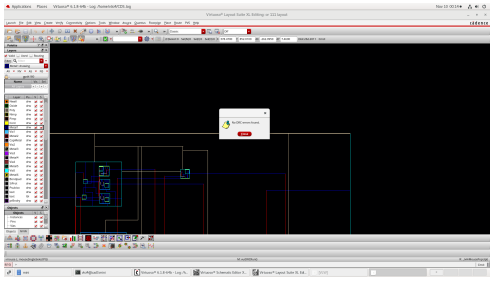
(b) AND



(c) OR



(d) XNOR



(e) 4-bit comparator

Figure 16. DRC results for all layouts

Table 7. DRC results

Layout	DRC result
Inverter	No DRC errors found
AND2	No DRC errors found
OR2	No DRC errors found
XNOR2	No DRC errors found
1-bit comparator	No DRC errors found
4-bit comparator	No DRC errors found

8 Results and Discussion

The design was carried through the complete full-custom flow, from the truth table to a DRC-clean layout of the 4-bit comparator. The main points are:

- The comparator is realised hierarchically from four basic cells, which were reused

to build the 1-bit comparator and the 4-bit comparator. This kept the number of different layouts small and the design easy to verify.

- Static CMOS logic gives full-swing outputs at a supply of 1.8 V.
- All six layouts (four basic cells, the 1-bit and the 4-bit comparator) pass the DRC without errors.
- Using the same transistor size for all devices simplified the design. The AND2, OR2 and XNOR2 cells use extra inverter stages, so they add area and delay compared with single-stage NAND2 or NOR2 implementations.
- The worst-case path of the 4-bit comparator runs from the least significant bit pair through the XNOR2 gate, the chain of AND2 gates and the final OR2 gate.

9 Conclusion and Future Work

9.1 Conclusion

A 4-bit magnitude comparator was designed in the gpdk180 180 nm CMOS technology using a full-custom flow in Cadence Virtuoso. Basic cells (INV, AND2, OR2 and XNOR2) were designed at transistor level and combined hierarchically into a 1-bit and then a 4-bit comparator. The schematics were created and simulated, the layouts of all cells and of both comparators were drawn manually, and all layouts were found to be free of design-rule violations. The project gave practical experience with the complete transistor-level design flow, from logic equations to physical layout.

9.2 Future Scope

- Single-stage NAND/NOR-based implementations to reduce the number of transistors and the delay.
- Transistor sizing optimisation (for example $W_p \approx 2W_n$) for balanced rise and fall times.
- Cascadable inputs (as in the 7485 comparator) and extension to 8-bit and 16-bit comparators.
- Migration to a smaller technology node such as 90 nm.